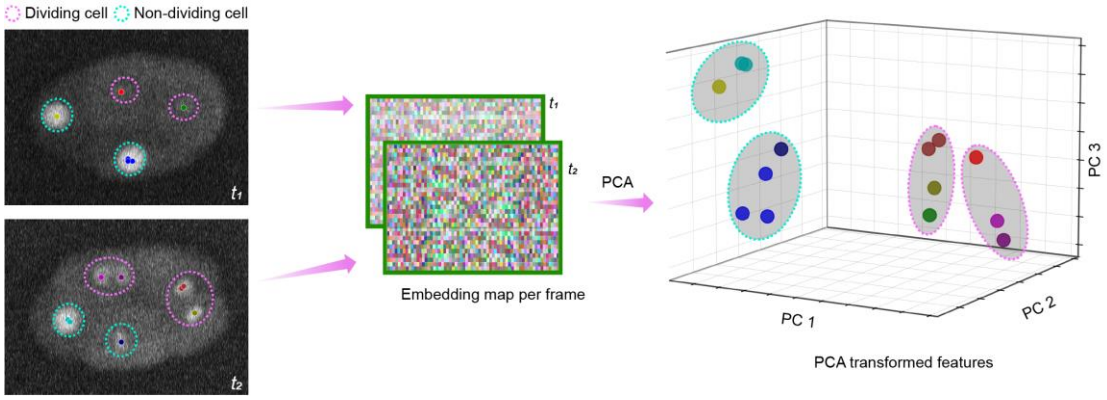


Supplementary Figure 3 | Clustering of feature vectors obtained by CELLECT via PCA across the adjacent frames. From 2 adjacent frames of input images, we extracted an embedding map for each frame and applied Principal Component Analysis (PCA) to reduce the dimensionality of feature vectors. In the resulting 3D PCA space, feature vectors corresponding to the same cell identity across the adjacent frames formed distinct clusters. Even dynamical changes during the division process, the feature distances of the daughter cells remained relatively close to that of the parent cell. By combining with Euclidean distances between cell centers, we inferred the potential division relationships between cells. The example shown here uses data from the Cell Tracking Challenge (CTC) Fluo-N3DH-CE dataset. The model used for feature extraction was trained on sequence #3 of the mskcc-confocal dataset.